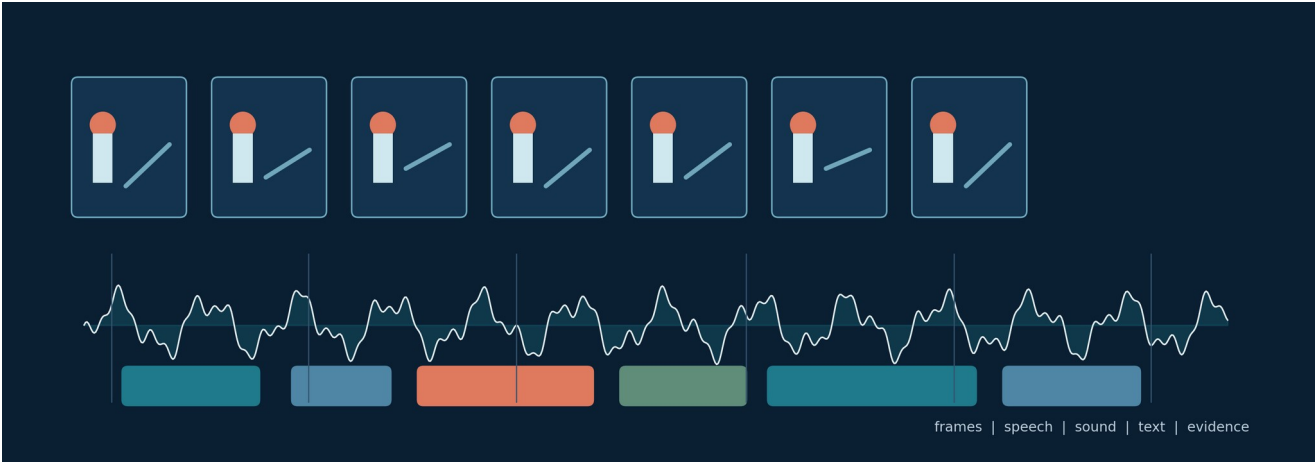


AV-Atlas

An Evidence-Grounded Hierarchical Audio-Visual Model for Comprehensive Long-Form Media Transcription

Working title | Architecture, training program, evaluation protocol, and governance plan



Concept motif: an audiovisual timeline transformed into timestamped evidence and language.

VERSION	1.0
DATE	13 July 2026
STATUS	Research concept - no experimental results claimed
SCOPE	Films, episodic media, and livestreams up to ten hours

Research-use notice. This document is a design proposal. Performance thresholds and compute figures are planning estimates to be tested. The legal discussion is risk framing, not legal advice.

Abstract

This paper proposes AV-Atlas, a custom audio-visual language model and associated training program for converting films, episodic media, and livestreams of up to ten hours into a detailed, time-aligned textual record. The target is not ordinary speech transcription or a short summary. It is comprehensive audiovisual transcription: dialogue, speakers, visible actions, objects, locations, camera changes, on-screen text, music, non-speech sound, uncertainty, and cross-scene continuity represented as auditable events. The central research problem is that raw long-form media is too information-dense for uniform frame sampling or a single monolithic context window, while unconstrained language generation can omit salient events or invent plausible details. AV-Atlas therefore adopts an evidence-first hybrid architecture. Specialist encoders perceive video, speech, speakers, acoustic events, and text; a continuous-time fusion model proposes event boundaries; bounded hierarchical memory carries shot-, scene-, chapter-, and title-level state; an entity registry preserves identity; and a structured decoder writes a canonical evidence ledger before producing prose. Every generated claim is linked to a time span and source evidence and can be accepted, revised, or withheld by a verifier. The deployment protocol combines a causal live pass with a retrospective correction pass. The recommended first implementation initializes from permissively licensed open weights but trains proprietary temporal fusion, memory, event, decoder, and verification modules. A staged curriculum progresses from seconds to ten hours, with explicit losses for temporal localization, coverage, grounding, consistency, and confidence calibration. The paper defines data-governance tiers, annotation ontology, evaluation metrics, ablation studies, compute envelopes, and a twelve-month research roadmap. It also identifies copyright, privacy, identity, prompt-injection, memorization, and benchmark-leakage risks. The proposal is falsifiable: success requires materially better supported-claim precision and salient-event recall than direct-prompt and modular baselines under a fixed compute budget.

Keywords: long-video understanding; dense video captioning; audio-visual transcription; streaming multimodal models; temporal grounding; hierarchical memory; evidence verification; dataset governance

PRIMARY RECOMMENDATION

Do not begin by pretraining every foundation component from random initialization. Build a genuinely custom model around trainable temporal fusion, event detection, hierarchical memory, entity continuity, structured decoding, and evidence verification, while initially reusing audited open encoders. Full video-foundation pretraining becomes an optional later phase after the task, data rights, and evaluation signal are proven.

Table 1. Concept at a glance.

Decision	Proposed position	Why it matters
Task	Comprehensive audiovisual transcription, not summarization	The output must preserve dialogue, visible events, sound, text, timing, identity, and uncertainty at multiple levels.
Architecture	Hybrid evidence-first model	It combines learnable end-to-end fusion with observability and specialist supervision.
Long-form strategy	Bounded hierarchical memory plus external evidence retrieval	Ten-hour inputs should not require all raw tokens to remain in the language model context.
Generation contract	Canonical JSONL evidence ledger before prose	Structured records enable validation, search, revision, and reproducible exports.
Deployment	Streaming provisional pass plus retrospective final pass	Live output and archival completeness are different operating points.
Data policy	Licensed/public/consented training data; private media defaults to inference or evaluation	Possession of a copy is not, by itself, a complete training-rights determination.
First research target	3-8B-class system with mostly frozen encoders	It is large enough to test the research hypothesis without paying foundation-model scale costs immediately.

Paper map

Part	Question answered
Sections 1-3	What is the task, why are existing approaches insufficient, and what is the research gap?
Sections 4-5	Which hypotheses will be tested, and what model architecture is proposed?

Part	Question answered
Sections 6-7	What data, annotation, governance, objectives, and curriculum are required?
Sections 8-9	How will the system be evaluated and what compute/infrastructure envelope is credible?
Sections 10-12	How should the work be staged, governed, de-risked, and published?
Appendices	What event schema, experiment matrix, and release checklist should be implemented?

1. Introduction

1.1 Motivation

Long-form media contains several synchronized information channels: images and motion; dialogue and speaker turns; music, effects, and ambient sound; subtitles, signs, interfaces, and credits; editing grammar; and narrative state accumulated over minutes or hours. A conventional transcript preserves only a portion of that evidence. A conventional summary discards most of it. The desired system must instead produce a dense but navigable textual representation that can support accessibility, archival search, research, compliance review, media analysis, and question answering.

Dense video captioning established the joint problem of detecting temporally localized events and describing them in language, with ActivityNet Captions providing approximately 20,000 videos, 849 hours, and 100,000 timestamped descriptions [1]. Streaming Dense Video Captioning later demonstrated fixed-size memory and online event generation for arbitrarily long streams [2]. These works provide important foundations, but the target here is broader: film-like continuity, accurate dialogue, non-speech audio, on-screen text, identity persistence, evidence traceability, and ten-hour operation.

Current long-video multimodal models show that large context windows and compressed video tokens can extend input duration. VideoPrism was pretrained on 36 million high-quality video-caption pairs plus 582 million text-linked clips [3]; InternVideo2.5 uses adaptive hierarchical token compression for long and detailed context [4]; LongVILA demonstrates long-context training and multi-thousand-frame inputs [5]; VideoTree organizes long video into an adaptive hierarchy [6]; and Qwen3-VL adds explicit timestamp alignment and a native long multimodal context [7]. These advances are useful starting points, but none alone guarantees comprehensive, evidence-grounded transcription of a ten-hour source.

RESEARCH GAP

The field optimizes mostly for question answering, retrieval, short captions, or coarse summaries. Our target optimizes a different frontier: high event coverage and low unsupported-claim rate over a very long timeline, while retaining auditability and the ability to revise earlier interpretations.

1.2 Proposed task: Comprehensive Audiovisual Transcription

We define Comprehensive Audiovisual Transcription (CAVT) as the transformation of a time-indexed audiovisual source into an ordered, multi-resolution event ledger. Each event record contains at minimum a start and end time, one or more atomic claims, participating entities, modality-specific evidence references, confidence values, and a revision state. Dialogue is represented verbatim only from an authorized subtitle or ASR channel; visual and acoustic descriptions are generated as evidence-grounded claims rather than as unconstrained narrative.

$e_i = (t_{start}, t_{end}, claims, entities, evidence_refs, confidence, revision_state)$

A canonical event is an auditable record, not merely a sentence.

The optimization target is deliberately multi-objective. The system should maximize coverage of salient observable events while minimizing unsupported claims, boundary error, identity switches, chronological contradictions, and compute. No single scalar metric fully captures this; the evaluation protocol therefore separates claim precision from event recall and adds evidence and temporal diagnostics.

maximize Coverage - alpha*UnsupportedClaims - beta*BoundaryError - gamma*IdentitySwitches - delta*Compute

Weights are selected on a validation set and reported with sensitivity analyses.

1.3 Claimed contributions of the proposed research

1. A formal task and output contract for evidence-linked, multi-resolution audiovisual transcription of media lasting seconds to ten hours.
2. An adaptive multi-rate perception front end that spends visual compute at cuts, motion bursts, speaker changes, acoustic novelty, and text appearance rather than at a uniform rate.
3. A bounded dual-memory architecture combining recurrent sensory state with hierarchical event, scene, chapter, and title memory.
4. An entity registry that links face, voice, appearance, object, and location evidence without forcing premature naming.
5. A canonical evidence ledger and verifier that require each generated claim to retrieve and survive checking against a precise source window.
6. A two-pass live/retrospective operating protocol and a training curriculum that scales duration only after local perception and grounding are stable.
7. A comprehensive evaluation protocol centered on supported-claim precision, salient-event recall, temporal accuracy, entity consistency, revision stability, and human audit.

2. Scope, requirements, and system boundaries

2.1 Target sources and output products

Table 2. Input scope and required behavior.

Input class	Typical properties	Required behavior
Films	90-240 minutes; cinematic editing; dialogue, score, effects, credits	Preserve narrative continuity, recurring characters, scene changes, visual action, dialogue, and music/effects without confusing inference with observation.
Episodes	20-90 minutes; recurring cast and locations; recaps and credits	Use a title-specific entity registry and distinguish recap/replayed material from current chronology.
Livestreams	1-10 hours; variable quality; long idle periods; overlays and chat	Produce low-latency provisional events with bounded memory, resilient chunking, and a later correction pass.
Recorded talks / gameplay / sports-like media	Dense speech, UI text, rapid action, repeated layouts	Adapt sampling to motion and text changes; keep transcript, visual state, and sound events synchronized.
Multi-track files	Several audio/subtitle languages, commentary, accessibility tracks	Inventory tracks, retain provenance, and select or combine them according to the task configuration.

The canonical output is machine-readable JSONL. Derived products include a word-aligned transcript (VTT/SRT), dense audiovisual timeline, scene narratives, chapter summaries, entity and location indexes, searchable embeddings, and direct links from each record to the source timecode. The model must be able to emit a minimal live view and a richer archival view from the same ledger.

2.2 Functional requirements

Table 3. Core functional requirements.

ID	Requirement	Verification idea
R1	Timestamp all atomic events and speech at the finest reliable resolution.	Compare boundaries against adjudicated intervals and word alignments.
R2	Describe visible actions, state changes, relationships, settings, and camera edits.	Human claim-level support and salient-event recall.
R3	Transcribe speech and associate speakers without allowing the visual decoder to invent quotations.	WER, diarization error, and quote-source audit.
R4	Describe non-speech audio and music cues with uncertainty.	Audio event F1/mAP plus human support judgments.
R5	Read on-screen text and preserve source/language metadata.	OCR CER/WER on text-bearing frames.
R6	Maintain entity continuity across cuts, costumes, lighting changes, and long absences.	Entity switch rate and identity-link precision.
R7	Process ten-hour media with bounded accelerator memory.	Peak-memory profile as duration increases.

ID	Requirement	Verification idea
R8	Provide provisional live records and final corrected records.	Latency, revision count, and final-delta analysis.
R9	Link every externally visible claim to source evidence and confidence.	Evidence retrieval precision/recall and schema validation.
R10	Abstain or mark ambiguity rather than smoothing over missing evidence.	Calibration, selective accuracy, and unsupported-claim rate.

2.3 Non-goals for the first research release

- Perfect frame-by-frame description of every pixel change. The objective is comprehensive semantic events, with denser re-analysis where the sampler detects uncertainty or fast change.
- Unrestricted real-person identification. Entity IDs are evidence clusters; names are attached only through an authorized registry or explicit human confirmation.
- Replacing the original media. The textual representation is an index and analytical layer, not a lossless substitute.
- Training on every personally owned or accessible title by default. Rights and privacy review govern whether a source is eligible for training, evaluation, or inference only.
- A single model checkpoint that simultaneously optimizes maximum live speed and maximum archival detail. The same architecture may expose separate operating profiles.

2.4 Architectural strategy decision

Table 4. Architectural strategy trade-off.

Strategy	Advantages	Limitations	Decision
Monolithic model from scratch	Clean end-to-end objective; complete architectural control.	Requires foundation-scale data/compute; weak observability; high legal and data risk before task value is proven.	Defer.
Loose modular pipeline	Fastest prototype; each component measurable; easy substitution.	Errors compound; modules optimize incompatible representations; narrative continuity is brittle.	Use as baseline.
Hybrid custom model	Own trainable fusion, memory, event, decoder, and verifier; specialist encoders provide strong initialization.	More engineering than a prompt pipeline; still requires license and dependency audit.	Recommended phase 1.

3. Related work and gap analysis

3.1 Dense captioning and streaming description

ActivityNet Captions formalized the joint detection-and-description problem and remains useful for event localization and caption metrics [1]. Streaming Dense Video Captioning adds a fixed-size clustering memory and streaming decoding, showing that event descriptions need not wait for the end of a video [2]. These systems are closest to the desired output form, yet their benchmark captions are usually sparse relative to a film transcript, and they do not provide a complete multimodal evidence ledger.

The proposed work extends dense captioning in four directions: multiple synchronized modalities; persistent entities; hierarchical long-term memory; and claim-level evidence verification. It also changes the evaluation emphasis. Caption similarity alone can reward fluent paraphrase while hiding omissions or unsupported details. CAVT must separately measure what was covered and whether each claim was supported.

3.2 Video foundations, long context, and hierarchical representations

VideoPrism illustrates the scale of modern video pretraining: tens of millions of curated video-caption pairs and hundreds of millions of noisy text-linked clips [3]. This is evidence against making full foundation pretraining the first milestone. InternVideo2.5 shows the value of adaptive hierarchical token compression for preserving both long context and fine detail [4]. LongVILA shows that long-video training is a systems problem as well as a modeling problem, reporting two-million-token training on 256 GPUs and extension from a handful of frames to thousands [5]. VideoTree uses a query-adaptive tree to represent long videos [6], and Qwen3-VL adds explicit timestamp alignment and long interleaved multimodal context [7].

These methods motivate hierarchical compression and timestamp-aware representations, but most are optimized for downstream question answering or instruction following. AV-Atlas instead keeps an always-on event ledger. The hierarchy is not built only after a query arrives; it is continuously updated, and each summary node retains links to the lower-level events from which it was formed.

3.3 Speech, diarization, sound, and text

Whisper demonstrates the robustness gained from 680,000 hours of weakly supervised multilingual speech data [8]. WhisperX adds voice-activity segmentation and forced alignment for accurate word timestamps, addressing drift and long-form processing issues [9]. pyannote.audio provides neural building blocks for speaker diarization [10]. General acoustic encoders such as BEATs [11] and language-audio models such as CLAP [12] provide complementary representations for non-speech sound and music. Ego4D contributes thousands of hours of consented egocentric video and audio with privacy-oriented collection practices [13], while MAD grounds more than 384,000 audio-description sentences in over 1,200 hours of film [14].

A key design choice is to supervise these channels separately before fusion. The language decoder should not be asked to reconstruct exact dialogue from sparse visual tokens, and an ASR model should not be asked to describe a camera move. Cross-modal fusion occurs at the event level after each branch has preserved its own timing and confidence.

3.4 Benchmarks and known reliability gaps

Video-MME covers diverse videos from seconds to an hour and explicitly evaluates video with audio and subtitles [16]. LongVideoBench provides interleaved video-language inputs up to an hour [17], while LVBench targets extreme long-video comprehension [18]. These benchmarks are useful but are dominated by question-answer accuracy. Work on temporal consistency shows that Video-LLMs can change their grounding under modest perturbations and that explicit temporal verification improves robustness [19]. VideoZeroBench advances evidence-based evaluation by pairing questions with temporal intervals and spatial boxes [20].

AV-Atlas adopts the spirit of evidence verification but applies it to open-ended generation. The model must not merely answer a question correctly; it must decide which observable events deserve records, locate them, describe them, and expose the evidence for each assertion. This creates a more demanding precision-recall problem.

Table 5. Related-work gap analysis.

Research line	What it solves	Remaining gap for AV-Atlas
Dense captioning [1-2]	Localized event sentences and streaming output.	Sparse modality coverage; limited persistent identity and audit trail.
Long-video MLLMs [4-7]	Large context, token compression, long-horizon QA.	Context length alone does not guarantee event completeness or grounded generation.
Speech/audio systems [8-12]	Accurate transcript, diarization, and acoustic semantics.	Need synchronized fusion, shared events, and film/stream continuity.
Long-video benchmarks [16-20]	QA, retrieval, temporal reasoning, evidence tests.	Few metrics for open-ended claim precision and event recall over ten hours.
Movie/egocentric data [13-15]	Narrations, grounded descriptions, daily activity diversity.	Licensing heterogeneity and insufficient gold annotation for the full proposed schema.

4. Research questions and falsifiable hypotheses

The project should be run as a sequence of falsifiable experiments rather than as a single large training job. The following questions define the minimum scientific contribution.

Table 6. Research questions, experiments, and falsification criteria.

RQ	Hypothesis	Primary experiment	Falsifier
RQ1: Memory	A bounded hierarchical memory preserves more salient events and entity state than a flat fixed-token context at equal memory.	Train matched models with flat, recurrent, and hierarchical memory; evaluate from 10 minutes to 10 hours.	No material recall/consistency gain, or gain disappears at matched compute.
RQ2: Sampling	Adaptive multi-rate sampling improves fast-event recall without increasing average visual tokens.	Compare uniform sampling with cut-, motion-, audio-, and uncertainty-triggered bursts.	Uniform sampling matches event recall and timing at equal token budget.

RQ	Hypothesis	Primary experiment	Falsifier
RQ3: Modality supervision	Independent ASR, acoustic, OCR, and visual losses reduce hallucination after fusion.	Ablate branch-specific supervision and audit unsupported claims by modality.	Joint-only training is equally accurate and more efficient.
RQ4: Evidence ledger	Forcing claim-evidence links improves factual precision and human trust.	Compare free-form decoder with ledger-constrained decoder and verifier.	No supported-claim improvement, or excessive abstention destroys coverage.
RQ5: Entity registry	Persistent multimodal entity memory lowers identity switches over long gaps.	Measure continuity across scene cuts, costume changes, and recurring speakers.	A simple text summary memory performs equally well.
RQ6: Two-pass operation	A retrospective pass recovers omissions and resolves ambiguities without large latency or contradiction costs.	Compare live-only, offline-only, and live-plus-retrospective outputs.	Second pass adds verbosity but not gold event recall or precision.
RQ7: Curriculum	Duration curriculum is more stable and data-efficient than immediate long-context training.	Match total tokens across curricula and compare convergence, memory use, and long-form score.	Direct long training converges as well or better at equal compute.

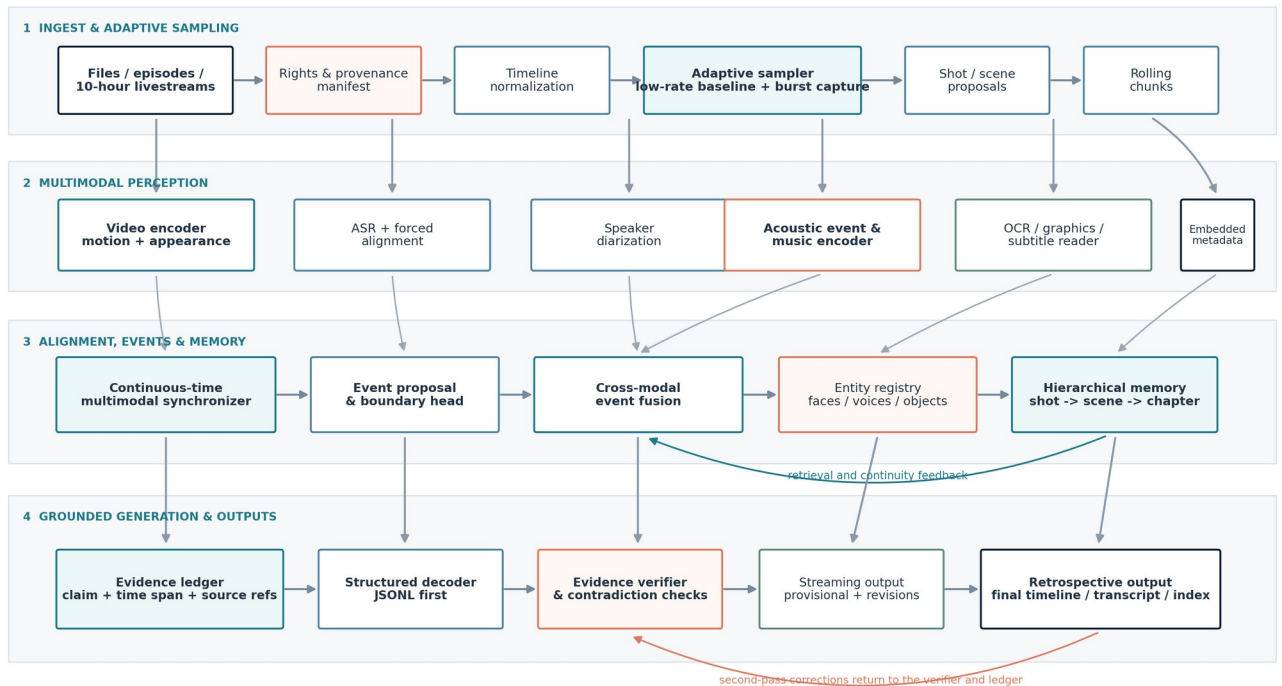
4.1 Expected scientific result

The strongest publishable result would not be merely a larger model. It would be evidence that an explicit event ledger, hierarchical memory, and claim verification shift the long-video precision-recall frontier. A credible negative result is also valuable: for example, if adaptive sampling or an entity registry provides no benefit at matched compute, that component should be removed rather than defended by intuition.

5. Proposed AV-Atlas architecture

AV-Atlas proposed architecture

A hybrid, evidence-first model for streaming and retrospective long-form audiovisual transcription



Design principle: generation is downstream of timestamped evidence, not a substitute for perception.

Figure 1. Proposed AV-Atlas architecture. All user-visible language is downstream of timestamped multimodal evidence.

5.1 Rights-aware ingest and timeline normalization

Every source enters through a provenance manifest containing a stable content ID, checksum, acquisition basis, rights category, permitted uses, retention policy, track inventory, language, frame rate, time base, and transformations. FFmpeg/ffprobe-class tooling can normalize containers and expose audio, video, subtitle, chapter, and metadata streams [29]. The model never treats a file path as sufficient provenance.

For livestreams, the ingest layer writes rolling, independently decodable chunks while maintaining a monotonic global time. A five- to fifteen-second overlap protects against cut speech and events at chunk boundaries. The exact chunk duration is an operational parameter; two to five minutes is a reasonable initial range because it balances retry cost, local context, and parallelism.

5.2 Adaptive multi-rate perception

Uniform one-frame-per-second sampling is efficient but can miss rapid action, brief text, and fast edits. AV-Atlas maintains a low-rate baseline and allocates burst frames when one or more novelty signals fire: shot boundary, optical-flow change, object/hand motion, speaker transition, acoustic onset, OCR appearance, decoder uncertainty, or explicit user interest. Static periods are compressed aggressively; fast sequences retain short high-rate windows.

The sampler is trainable. A teacher operating on dense frames can score which frames support the gold event description. A lightweight policy predicts those scores from cheaper features and is optimized under a token budget. During evaluation, all methods receive the same average number of visual tokens so any gain can be attributed to allocation rather than extra compute.

5.3 Modality-specific encoders

Table 7. Perception branches and initial implementations.

Branch	Input	Output tokens / records	Initial implementation
Visual	Frames and short clips at adaptive rates.	Appearance, motion, spatial relations, shot features, object tracks.	Video-native encoder or strong open VLM vision tower; preserve timestamp per token.
Speech	Dialogue-oriented audio mix.	Word tokens, timings, language, confidence.	Whisper/WhisperX-class ASR and forced alignment [8-9].
Speaker	Speech embeddings and turn segments.	Anonymous speaker clusters and overlap estimates.	pyannote-class diarization with title-specific adaptation [10].
Acoustic	Full audio mix, optionally stems.	Sound-event, ambience, music, and onset embeddings.	BEATs/CLAP-class encoder plus event head [11-12].
Text	Selected frames, subtitles, graphics, metadata.	OCR spans, language, geometry, track provenance.	Dedicated OCR plus subtitle parser; VLM reader as fallback.
Metadata	Timecode, chapters, codec/track information.	Absolute timing and source descriptors.	Deterministic parser; never generated by the LLM.

The first training stage freezes most large encoders and learns projection layers into a shared event space. Selective unfreezing follows only when branch-level validation shows a domain gap. This protects the project from catastrophic forgetting and makes attribution of improvements clearer.

5.4 Continuous-time synchronizer

All tokens are assigned absolute start/end times before fusion. The synchronizer combines learned modality embeddings with continuous time features rather than relying only on token position. Cross-attention is restricted by temporal neighborhoods at the local level but can retrieve remote events through memory. The design must represent uncertainty: an OCR token visible from 12.0-12.4 seconds and a sound whose onset is uncertain by 300 milliseconds should not be collapsed into identical point timestamps.

Training includes deliberately misaligned negatives: shifted subtitles, swapped audio, repeated frames, delayed effects, and dubbed speech. The synchronizer learns to prefer the correct alignment and to flag cases where modalities disagree rather than silently forcing a match.

5.5 Event proposal and fusion

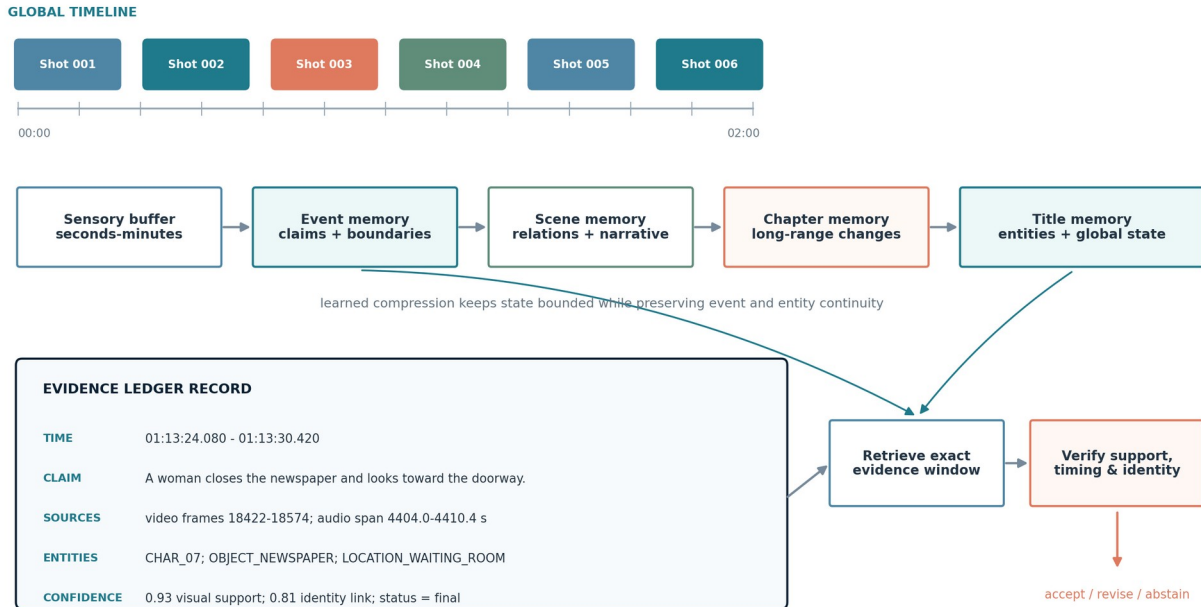
A DETR-like set-prediction head [23] proposes event intervals and categories without requiring a fixed number of events. Event queries attend to synchronized modality tokens and regress start/end times, salience, modality participation, and confidence. Bipartite matching aligns proposals to annotated events. A separate change-point head detects scene and state transitions that are not well represented as short actions.

The event-fusion transformer produces an event state with atomic predicates rather than a polished sentence: participant, action or state, object, location, manner, visual evidence, speech span, acoustic evidence, text evidence, and uncertainty. The structured decoder later lexicalizes this state. Keeping an intermediate semantic form makes it possible to test whether an error originated in perception, fusion, or language generation.

5.6 Dual memory and entity continuity

Bounded memory with an auditable evidence ledger

Raw media stays external; the model carries forward compressed state and retrieves evidence when a claim is generated.



The ledger supports traceability, incremental revisions, human audit, and retrieval-based question answering without replaying all ten hours.

Figure 2. Bounded hierarchical memory and the evidence-ledger verification loop.

The short-term sensory memory is a fixed-size recurrent buffer containing recent multimodal tokens and unresolved proposals. The long-term episodic memory is hierarchical: atomic events are compressed into scene nodes; scene nodes into chapter nodes; chapter nodes into title-level state. Each summary node stores references to its children, enabling exact evidence retrieval. Compression is learned with reconstruction, retrieval, and consistency objectives, not by free-form summarization alone.

The entity registry is a parallel memory keyed by anonymous IDs. It stores face and body embeddings, voice embeddings, clothing/appearance descriptors, names from authorized metadata, co-occurrence relations, and confidence. Identity is a link hypothesis, not a fact. The model may state “the same speaker as SPEAKER_03” at high confidence while withholding a name. Human confirmation can promote a cluster to a named character and retroactively update the ledger without rewriting the evidence.

5.7 Evidence ledger, structured decoder, and verifier

The decoder is constrained to emit a schema-valid event record before any narrative. Grammar-constrained decoding or a typed intermediate representation prevents missing fields and malformed timestamps. Prose products are deterministic or model-assisted views over the ledger. This separation makes the source of a quotation explicit and lets users choose a concise or dense rendering without rerunning perception.

The verifier retrieves the exact source window referenced by each claim and scores visual support, audio support, text support, temporal fit, entity consistency, and contradiction with prior state. Unsupported modifiers can be removed while the rest of a claim is retained. Low-confidence claims are marked uncertain or withheld. Verification tuning should include hard negatives that are plausible in the scene but absent from the evidence, because ordinary next-token training rewards plausibility more than observability.

5.8 Streaming and retrospective modes

In streaming mode, AV-Atlas emits provisional events after a bounded delay. Records carry stable IDs and revision numbers. The model may leave a speaker anonymous, defer a scene label, or mark an event open until its end boundary is observed. In retrospective mode, the system revisits uncertain windows, resolves cross-chunk entities, removes overlap duplicates, identifies missed events, and reconciles later context with earlier claims.

The final ledger is not necessarily identical to the live transcript, and that difference should be measured. Revision stability is a first-class metric: a useful live system should improve when corrected without continuously rewriting already stable history.

5.9 Model scale and ownership interpretation

“Our own model” is interpreted here as a distinct architecture, training run, weight set, data recipe, evaluation suite, and deployment stack under our control. It does not require random initialization of every encoder in the first phase. Qwen3-VL, for example, provides long context and timestamp-aware video representations, and its official repository carries an Apache-2.0 license [7,28]; however, every selected checkpoint and dependency must be audited separately. WhisperX code is permissive, but alignment and diarization dependencies can have different terms. A model bill of materials is therefore mandatory.

PHASE BOUNDARY

Only after the hybrid system demonstrates measurable value should the project consider pretraining a proprietary video or audio foundation encoder. VideoPrism-scale training data [3] indicates that this is a separate program, not an incremental fine-tune.

6. Data, annotation, rights, and governance

6.1 Data tiers

Table 8. Proposed data-rights tiers.

Tier	Examples	Permitted default use	Controls
A - Clear training rights	Public-domain, suitable Creative Commons, commissioned, directly licensed, or internally produced media.	Training, validation, and testing according to the specific license/contract.	Store license text, attribution, expiry, territory, and permitted model uses.
B - Research datasets	ActivityNet Captions, YouCook2, Ego4D, MAD, benchmark data.	Only as allowed by dataset terms; some may be research-only or have redistribution limits.	Dataset-specific review, access logging, no assumption that paper availability equals commercial rights.
C - Private/user-accessible media	Personally owned discs/files, subscription media, private episodes, recorded streams.	Inference and private evaluation by default; training only after a documented rights decision.	No redistribution, encrypted storage, retention limit, opt-out/deletion path.
D - Synthetic/controlled	Rendered scenes, staged recordings, synthetic overlays, generated sound mixtures.	Training and adversarial testing under generator/asset terms.	Track source assets; avoid synthetic labels that encode teacher errors as truth.
E - Excluded/quarantined	Unclear provenance, rights reservation, sensitive biometric material without basis, leaked or unlawfully obtained content.	No training until resolved.	Quarantine store and review workflow.

For an EU deployment, Directive (EU) 2019/790 Article 4 conditions the general text-and-data-mining exception on lawful access and the absence of an effective rights reservation [24]. German Section 44b UrhG similarly permits reproductions of lawfully accessible works for text and data mining only where use has not been reserved, while Section 60d provides a separate scientific-research framework [25]. These rules are context-sensitive. Possessing a disc or subscription is therefore relevant to access but is not, by itself, a complete conclusion that model training is permitted. Obtain jurisdiction-specific counsel before moving Tier C media into training.

For a general-purpose AI model placed on the EU market, Article 53 of the AI Act includes copyright-compliance and training-content-summary obligations [26]. Even if the initial system is private or specialized, the project should build provenance and summary capabilities from day one because retrofitting them after training is difficult. Privacy governance must also consider the GDPR when face, voice, account, or chat data relates to identifiable people [27].

6.2 Training corpus plan

Table 9. Initial corpus planning targets. These are design estimates, not collected data.

Corpus slice	Planning target	Annotation density	Purpose
Weakly labeled licensed corpus	2,000-5,000 hours initially; expand only after pilot.	Subtitles/ASR, shot boundaries, metadata, teacher event candidates.	Representation alignment and domain breadth.
Dense event corpus	150-300 hours.	Atomic events, boundaries, modalities, evidence, uncertainty, entities.	Core CAVT supervision and verifier training.

Corpus slice	Planning target	Annotation density	Purpose
Long-form curriculum corpus	500-1,500 hours across episodes, films, talks, and streams.	Scene/chapter structure, entity links, sparse gold plus dense teacher labels.	Memory, continuity, and duration scaling.
Adjudicated gold test	40-80 hours, isolated from model and prompt development.	Double annotation, adjudication, exact evidence links.	End-to-end scientific claims.
Ten-hour stress set	At least six diverse 8-10 hour recordings.	Event samples throughout, full transcript, targeted dense windows.	Bounded memory, live latency, revision stability, drift.

6.3 Annotation ontology

Annotations are event-centered. An annotator first marks a minimal observable interval, then records participant IDs, actions or states, objects, location, spatial relations, dialogue linkage, non-speech sound, music, on-screen text, camera/edit information, salience, uncertainty, and evidence frames/audio. Events can be nested: a two-second glance may sit inside a thirty-second conversation and a five-minute scene. The ontology should allow “unknown” and “not visible” rather than pressuring annotators to infer.

Table 10. Core annotation ontology.

Field group	Examples	Annotation rule
Temporal	start, end, open/closed, boundary confidence	Use the shortest interval that contains the observable event; record tolerance for gradual changes.
Participants	anonymous person/speaker IDs, groups, off-screen speaker	Link only when evidence supports continuity; naming is a separate registry action.
Visual predicate	enters, picks up, looks toward, remains seated, light turns off	Prefer atomic changes; avoid motives unless explicitly expressed.
Audio predicate	door slam, applause, siren, tense instrumental music	Describe audible evidence; distinguish source-visible, off-screen, and uncertain.
Text	subtitle, sign, UI, lower third, credits	Store exact OCR string, geometry, language, and source type where permitted.
Camera/edit	cut, dissolve, pan, close-up, split screen	Annotate when it changes interpretation or searchability.
Evidence/uncertainty	frame IDs, audio span, confidence, alternatives	Every claim must be traceable; ambiguity is represented explicitly.

6.4 Annotation workflow and quality control

8. Run deterministic extraction and a teacher ensemble to propose shots, speech, speaker turns, OCR, sounds, and candidate events.
9. Have a primary annotator correct boundaries and claims in a synchronized media tool.
10. Have a second annotator verify salient events, evidence, and identity links without seeing model confidence.
11. Adjudicate disagreements on a gold subset and update the ontology guide with concrete examples.
12. Use active learning to prioritize high-uncertainty, high-novelty, and high-disagreement windows.
13. Fingerprint media and annotations to prevent train/test leakage, near-duplicate clips, and repeated versions of the same title.

Inter-annotator agreement must be reported separately for boundaries, event semantics, salience, and entity links. A single agreement number would hide the fact that two annotators can agree on what happened but differ by several seconds on where a gradual event begins.

6.5 Provenance and model bill of materials

Each training example should carry a machine-readable lineage: source ID; rights tier; license or contract; source hash; extraction commands; teacher model versions; human annotation versions; transformations; split assignment; and deletion/expiry status. The same discipline applies to pretrained weights, code, tokenizers, alignment models, and evaluation data. A component is not approved merely because its top-level repository has a permissive license; dependent checkpoints and gated models must be reviewed.

7. Training program and objectives

Training curriculum and deployment loop

Increase duration only after perception, temporal grounding, and structured output are stable at the previous scale.

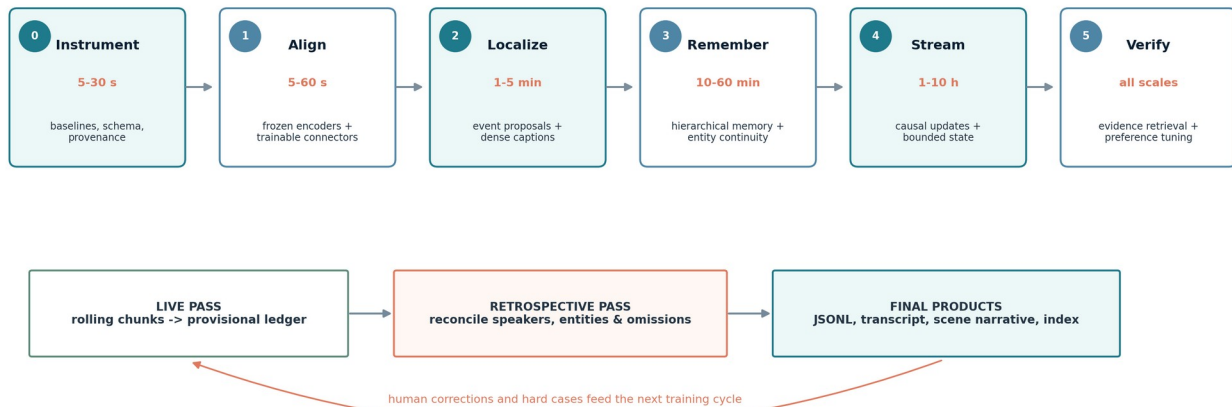


Figure 3. Proposed duration curriculum and the live/retrospective deployment loop.

7.1 Stage 0 - Instrumentation and baselines

Before model training, implement reproducible ingest, the event schema, a feature store, a media viewer, annotation tools, and benchmark harness. Establish three baselines: direct prompting of a strong open video-language model on uniform frames; a modular pipeline combining ASR, OCR, audio tagging, and text fusion; and a published dense/streaming captioning approach where feasible. All later gains are reported against matched token and compute budgets.

7.2 Stage 1 - Connector and time alignment training

Freeze the large visual, speech, speaker, audio, and OCR encoders. Train modality projectors, continuous-time embeddings, and local cross-modal alignment on five- to sixty-second clips. Objectives include ASR/word alignment, audio-text contrastive loss, OCR recognition, modality correspondence, and hard-negative synchronization. This stage should expose whether the selected encoders cover the domain before long-context complexity is introduced.

7.3 Stage 2 - Dense event localization and description

Train event queries and structured semantic records on one- to five-minute windows. Use bipartite matching between proposals and gold events, with boundary, eventness, salience, and predicate losses. The decoder is trained to emit schema-valid JSON and then a textual description. Dialogue spans are copied or referenced from the speech channel rather than regenerated from memory.

7.4 Stage 3 - Hierarchical memory and entity registry

Extend windows through ten, thirty, and sixty minutes. Train memory compression by requiring scene/chapter states to retrieve their child events, answer state queries, reconstruct entity changes, and remain consistent under shuffled irrelevant segments. Entity training includes cross-cut positives, same-actor/different-character hard negatives where licensing permits, same-character/different-costume positives, and voice/face disagreement cases.

7.5 Stage 4 - Ten-hour streaming adaptation

Switch the sensory path to causal attention and fixed memory. Train on rolling chunks with state carryover, connection interruption, missing frames, clock drift, and duplicated overlap. Losses apply both to per-chunk provisional records and to the merged global ledger. The model learns explicit open events and revision operations rather than pretending that future context is already known.

7.6 Stage 5 - Verification and preference optimization

Create positive and negative claim-evidence pairs. Negatives should include plausible but absent objects, wrong participants, nearby but incorrect times, dialogue attributed to the wrong speaker, sounds assigned to visible sources without evidence, and subtle chronology inversions. Train the verifier with classification, pairwise ranking, and calibration losses. Preference optimization can favor outputs that retain supported detail while deleting unsupported embellishment, avoiding the common failure mode where safety is achieved only by becoming vague.

7.7 Multi-task loss

$$\begin{aligned} L_{\text{total}} = & w_{\text{evt}} * L_{\text{event}} + w_{\text{cap}} * L_{\text{caption}} + w_{\text{asr}} * L_{\text{ASR}} + w_{\text{aud}} * L_{\text{audio}} + w_{\text{ocr}} * L_{\text{OCR}} + \\ & w_{\text{ent}} * L_{\text{entity}} + w_{\text{grd}} * L_{\text{ground}} + w_{\text{cov}} * L_{\text{coverage}} + w_{\text{con}} * L_{\text{consistency}} + \\ & w_{\text{cal}} * L_{\text{calibration}} \end{aligned}$$

Table 11. Proposed training objectives.

Term	Purpose	Example signal
L_event	Detect event existence, category, salience, and temporal boundaries.	Set matching, classification, boundary regression, temporal IoU.
L_caption	Generate structured predicates and faithful language.	Token loss conditioned on gold semantic event records.
L_ASR / L_audio / L_OCR	Preserve modality competence.	Word targets, sound labels/text contrast, OCR strings and geometry.
L_entity	Maintain stable anonymous identities and relations.	Contrastive links, cluster loss, cross-modal agreement.
L_ground	Attach each claim to supporting evidence.	Claim-window contrastive score and retrieved-evidence classification.
L_coverage	Penalize omission of annotated salient events.	Set recall surrogate and unmatched-gold penalty.
L_consistency	Keep descriptions compatible across scales and revisions.	Event-scene-chapter entailment and chronology constraints.
L_calibration	Make confidence and abstention meaningful.	Brier loss, expected calibration error, selective risk.

Loss weights should not be tuned only on aggregate language scores. A Pareto analysis should show how supported-claim precision trades against event recall and compute. Dynamic weighting may be useful, but fixed-weight baselines are necessary for interpretation.

7.8 Data augmentation and hard cases

- Temporal: speed changes, frozen frames, jump cuts, replayed scenes, missing chunks, shifted subtitles, and gradual fades.
- Visual: compression artifacts, low light, overlays, split screens, brief text, animation, and high-motion bursts.
- Audio: noise, reverberation, dubbing, overlapping speakers, music over dialogue, off-screen sound, and channel downmix variation.
- Cross-modal: swapped audio, contradictory subtitles, duplicated dialogue, and a visible action with delayed sound.
- Narrative: recurring locations, long entity absences, same actor in multiple roles, recaps, flashbacks, and footage-within-footage.
- Security: adversarial on-screen or spoken instructions that must be recorded as content, never executed as model control.

7.9 Parameter-efficient versus full training

The first prototype should train projectors, event heads, memory, registry, decoder adapters, and verifier while freezing most encoders and the lower language backbone. The core model then selectively unfreezes upper visual/time layers and language layers if validation identifies a bottleneck. Full end-to-end fine-tuning is a controlled experiment, not an assumption. Full encoder pretraining from scratch is a later program with a new data and compute budget.

8. Evaluation protocol

8.1 Evaluation philosophy

A fluent description is not sufficient evidence of understanding. Evaluation must decompose five questions: Did the model notice the event? Did it locate the event? Is the claim supported? Did it preserve the correct entity and chronology? Did it use resources efficiently? Automatic metrics are necessary for iteration, but claim-level human evaluation is the final arbiter for open-ended completeness.

8.2 Component metrics

Table 12. Component evaluation.

Component	Metrics	Failure diagnosed
Speech	WER/CER, word-boundary error, hallucinated-speech rate.	Misheard dialogue, drift, repetition, fabricated words.
Diarization	DER/JER, speaker-attribution F1, overlap accuracy.	Speaker switches, merged speakers, missed overlap.
OCR	CER/WER, detection recall, track/source accuracy.	Missed brief text, wrong reading, subtitle/UI confusion.
Acoustic events	mAP/F1, onset/offset error, source-visible accuracy.	Missed sound, wrong label, unsupported source attribution.
Event proposal	mAP at temporal IoU thresholds, boundary MAE, proposal recall.	Missed or over-segmented events.
Entity registry	Link precision/recall, switch rate per hour, naming precision.	Identity drift and false merging.
Schema	Validity, required-field completeness, timestamp monotonicity.	Operational failures independent of semantics.

8.3 End-to-end CAVT metrics

Table 13. End-to-end metrics.

Metric	Definition	Preferred direction
Supported-claim precision	Fraction of atomic generated claims judged supported by linked evidence.	Higher; primary trust metric.
Salient-event recall	Fraction of human-marked salient events represented by at least one correct record.	Higher; primary completeness metric.
Evidence precision/recall	Whether cited windows contain the claimed evidence and whether all needed evidence was retrieved.	Higher.
Boundary error	Absolute and relative difference between predicted and gold event intervals.	Lower.
Chronology violation rate	Contradictions in ordering, state transitions, or flashback labeling.	Lower.
Entity switch rate	Incorrect changes or merges of persistent entity IDs per hour.	Lower.
Coverage density curve	Event recall as output token budget increases.	Dominates baselines.
Selective risk	Unsupported-claim rate as low-confidence records are withheld.	Lower at useful coverage.
Revision stability	Fraction and magnitude of live records changed by the final pass.	Lower for stable records; corrections should improve accuracy.
Efficiency	Tokens, accelerator-seconds, peak memory, and energy proxy per input hour.	Lower at matched quality.

8.4 Benchmark suite

Public evaluation should include ActivityNet Captions and YouCook2 for dense events [1,15]; MAD for film audio-description grounding [14]; Video-MME for multimodal duration and audio/subtitle conditions [16]; LongVideoBench and LVBench for extended context [17-18]; temporal-consistency probes [19]; and evidence-focused VideoZeroBench [20]. AudioSet and VGGSound can evaluate general sound representations [21-22]. None fully measures CAVT, so the internal adjudicated AV-Atlas Gold set is essential and should be released in a rights-compatible form if possible.

8.5 Baselines and ablations

Table 14. Minimum baseline and ablation matrix.

Experiment	Change	Question
B0 Direct VLM	Uniform frames plus audio/transcript in one prompt.	How far does a strong general model go without the proposed machinery?
B1 Modular pipeline	Independent components merged by a text model.	What benefit comes from learned event-level fusion?
A1 No adaptive sampler	Uniform sampling at the same visual-token budget.	Does allocation improve fast-event recall?
A2 Flat memory	Replace hierarchy with equal-size flat cache.	Does hierarchy improve duration scaling?
A3 No entity registry	Use only generated text summaries.	Does multimodal entity state reduce switches?
A4 No evidence ledger	Generate free-form prose.	Does the ledger improve precision and auditability?
A5 No verifier	Accept first-pass records.	How much precision/calibration comes from verification?
A6 No retrospective pass	Use streaming output as final.	What is the value and cost of second-pass correction?
A7 No branch losses	Train only end-to-end caption loss.	Do modality-specific objectives reduce hallucination?

8.6 Human study

Human evaluation should use blinded side-by-side comparisons on matched source windows. Annotators score factual support, missed salient events, timing, speaker/character consistency, useful uncertainty, and readability. Each atomic claim receives supported / partly supported / unsupported / unverifiable labels, and each gold event receives covered / partially covered / missed labels. At least two annotators plus adjudication are required for the final test set.

For long media, do not ask a single evaluator to remember ten hours. Divide annotation into overlapping windows, then run a separate continuity pass over scene and chapter summaries. Sample throughout the duration, not only near the beginning, to reveal memory drift.

8.7 Provisional success gates

Table 15. Planning targets. These are proposed gates, not achieved results.

Gate	Target for research continuation	Interpretation
G1 Short-window validity	Schema error below 1%; supported-claim precision above 90% on adjudicated 1-5 minute clips.	The output contract and verifier are working.
G2 Dense event benefit	At least 10% relative improvement in salient-event recall over direct VLM at matched claim precision and token budget.	The architecture adds value beyond prompting.
G3 Duration scaling	No more than 15% relative degradation in precision/recall from 30 minutes to 2 hours; bounded peak memory.	Hierarchical memory is credible.
G4 Ten-hour streaming	Provisional latency under 60 seconds at the selected profile; no unbounded memory growth; successful recovery from chunk loss.	The live system is operational.
G5 Final correction	Retrospective pass improves event recall or precision without increasing chronology violations.	Two-pass design is justified.
G6 Governance	100% of training examples traceable to a rights/provenance record; deletion test demonstrated.	The corpus is operable and auditable.

9. Compute, systems, and infrastructure

9.1 Model variants and planning envelopes

Table 16. Order-of-magnitude planning estimates; benchmark on the selected hardware before budgeting.

Variant	Model plan	Accelerator envelope	Estimated training scale	Role
Prototype S	3-4B language backbone; frozen encoders; 0.3-0.8B trainable parameters.	8 x 80 GB-class GPUs.	2,000-5,000 GPU-hours.	Validate schema, event model, ledger, and short/medium memory.
Core M	Approximately 8B backbone; selective unfreezing; 1-2B trainable parameters.	16 x 80 GB-class GPUs.	10,000-30,000 GPU-hours.	Primary research model for films, episodes, and ten-hour streaming.
Research L	14-32B backbone or MoE; broader end-to-end tuning.	32-64 x 80 GB-class GPUs.	40,000-120,000+ GPU-hours.	Quality ceiling, scaling study, and teacher model.
Foundation phase	Proprietary video/audio pretraining from random or near-random initialization.	128-256+ accelerators plus dedicated storage/network.	Separate budget; potentially orders of magnitude larger.	Only after task and rights signal are proven.

The GPU-hour ranges are not quotes and should not be converted directly into cost without measuring sequence length, frame count, precision, checkpointing, sequence parallelism, data-loader efficiency, and utilization. Long-video training often becomes input/output and network bound before nominal floating-point capacity is exhausted.

9.2 Storage and data plane

Use immutable original storage, normalized media proxies, a timestamped feature store, annotation database, and versioned event ledgers. Raw media and derived features should have separate retention and access policies. Features must not be treated as legally or privacy-neutral; face and voice embeddings can remain identifying. Training jobs read manifests rather than crawling directories, which enables rights filtering, split enforcement, and deletion propagation.

The live path requires resilient chunk queues, idempotent processing, state checkpoints, and exactly-once ledger merge semantics. A ten-hour session must survive a worker restart without losing global time or duplicating already finalized records. The retrospective path can be asynchronous and use denser sampling, larger batches, and higher-quality verifier settings.

9.3 Reproducibility and observability

- Version every encoder, tokenizer, prompt template, schema, loss configuration, and sampler policy.
- Log per-modality token counts, selected frames, event proposals, memory writes/reads, verifier outcomes, and revisions.
- Store deterministic evaluation manifests and source hashes; prohibit test-set media or near-duplicates from teacher-label generation.
- Report quality against actual input duration and compute, not only model parameter count.
- Maintain a model/data bill of materials and a deletion/retraining impact map.

10. Twelve-month implementation roadmap

Table 17. Proposed twelve-month roadmap.

Months	Workstream	Deliverable	Decision gate
0-2	Governance, ingest, schema, annotation viewer, baselines.	Reproducible pipeline; rights manifest; 20-40 hour pilot set; B0/B1 results.	Proceed only if annotations and provenance are operational.
2-4	Connector alignment and short-window event model.	Prototype S on 5-60 second clips; valid JSONL; component dashboards.	G1 short-window validity.
4-6	Dense event proposals, evidence ledger, verifier.	1-5 minute CAVT model; hard-negative set; claim-level human study.	G2 dense-event benefit.
6-8	Hierarchical memory and entity registry.	30-120 minute model; continuity and memory ablations.	G3 duration scaling.
8-10	Streaming adaptation and operations.	Ten-hour provisional pipeline; restart/recovery tests; live latency profile.	G4 ten-hour streaming.

Months	Workstream	Deliverable	Decision gate
10-12	Retrospective correction, full benchmark, release package.	Core M report; gold evaluation; model/data cards; publication draft.	G5-G6 and release review.

10.1 Team shape

A compact research team should cover video-language modeling, speech/audio, distributed training, data/annotation systems, evaluation, and rights/privacy governance. The critical dependency is not only GPU access; it is the ability to build and adjudicate a high-quality event dataset. A suggested minimum is two multimodal researchers, one speech/audio researcher, two ML/systems engineers, one data/annotation lead, and fractional legal/privacy review, with annotator capacity scaled to the pilot results.

10.2 Milestone artifacts

14. Task specification, event schema, annotation guide, and provenance schema.
15. Reproducible baseline repository and evaluation harness.
16. Prototype S and Core M checkpoints with model cards.
17. Rights-filtered dataset manifests and data cards; no unapproved raw-media redistribution.
18. AV-Atlas Gold evaluation set or a releasable substitute with adjudicated evidence links.
19. Technical report including negative results, compute, ablations, and error taxonomy.
20. Deployment reference for live chunking, revision handling, and retrospective finalization.

11. Risks, ethics, security, and limitations

Table 18. Principal risks and controls.

Risk	Failure mode	Mitigation / test
Copyright and contract	Training sources are lawfully accessed but not authorized for the intended use; outputs reproduce scripts or distinctive expression.	Rights tiers, counsel review, reservations/terms capture, memorization audits, transcript access controls, source deletion workflow.
Privacy and identity	Face/voice embeddings or chat data identify people beyond the intended context.	Data minimization, pseudonymous entity IDs, access control, retention limits, DPIA where applicable, no automatic naming without basis.
Hallucination	Plausible actions, objects, motives, or sounds are added without evidence.	Ledger-first decoding, verifier hard negatives, claim precision metric, calibrated abstention.
Omission	The model produces a polished but incomplete narrative.	Coverage loss, event proposal recall, adaptive re-analysis, gold salient-event annotations.
Temporal drift	Timestamps and chronology degrade across chunks or long gaps.	Absolute time normalization, overlap reconciliation, temporal consistency tuning [19], stress tests.
Identity drift	Recurring people/speakers are split or merged.	Multimodal registry, confidence thresholds, manual confirmation, entity switch audit.
Prompt injection in content	On-screen text or spoken instructions influence system behavior.	Treat all media text/audio as quoted data; isolate control tokens; adversarial media tests.
Benchmark leakage	Training includes benchmark clips or near-duplicates.	Perceptual hashes, audio fingerprints, title/version matching, frozen test manifests.
Teacher-label bias	Synthetic labels become self-reinforcing and suppress novel events.	Human gold subset, teacher diversity, disagreement sampling, periodic label audits.
Resource intensity	Quality requires too many frames/tokens to be practical.	Matched-compute experiments, adaptive sampling, model variants, token/energy reporting.
Sensitive media	Violence, sexual content, minors, private conversations, or protected information appears in sources.	Restricted-access processing, policy-specific redaction, trained annotator support, content handling plan.
Overclaiming	Concept targets are presented as achieved performance.	Clearly separate hypotheses, targets, and measured results in every report and model card.

11.1 Memorization and output governance

Because films and episodes contain copyrighted dialogue, the system should separate authorized transcript storage from generated visual description. Evaluate exact and near-exact extraction under adversarial prompts, compare outputs to subtitles/scripts where legally available, and test whether rare training sequences can be reconstructed. Public products may need quote-length limits or access controls even when private analytical use is permitted. These are product and legal decisions, not only model metrics.

11.2 Limits of comprehensiveness

No practical system will guarantee that every semantically meaningful micro-event in ten hours has been captured. Visibility, compression, overlapping sound, ambiguity, cultural context, and annotation disagreement impose limits. AV-Atlas should report operating profiles and confidence rather than advertise lossless understanding. The scientific claim is an improved and measurable precision-recall frontier, not perfect perception.

11.3 Bias and accessibility

Evaluation should stratify by language, accent, skin tone, lighting, animation/live action, audio quality, hearing accessibility tracks, and caption availability. Film audio descriptions can provide valuable training signals but reflect particular narrative conventions and may omit details that a dense analytical transcript requires [14]. Accessibility users should participate in defining useful granularity, uncertainty wording, and navigation.

12. Expected contributions, release strategy, and conclusion

12.1 Expected research contributions

The project can make a meaningful contribution even without the largest model. The novel center is an evidence-grounded task formulation and architecture for comprehensive long-form generation. Publishable outcomes include a claim/evidence schema; bounded hierarchical memory; adaptive multimodal sampling; coverage and consistency objectives; a live-plus-retrospective protocol; a gold evaluation method; and a transparent error taxonomy.

12.2 Release strategy

Release in layers according to rights and risk: code for ingest, schema validation, metrics, and synthetic examples; model adapters or weights where component licenses allow; annotation guidelines; aggregate dataset statistics; and a rights-compatible evaluation subset. Raw copyrighted film or episode content should not be redistributed. The release should include model and data cards, known failures, intended uses, prohibited uses, compute, and training-content summary at the appropriate level.

12.3 Conclusion

AV-Atlas is technically feasible as a staged research program, but success depends less on placing ten hours into one context window than on disciplined representation. The proposed system turns raw media into timestamped, modality-specific evidence; fuses that evidence into atomic events; stores long-range state in a bounded hierarchy; and allows language only after claims can be traced and verified. The recommended first model is a hybrid custom architecture initialized from audited open components. It provides a credible path to proprietary weights and differentiated capability without prematurely undertaking foundation-scale pretraining. The decisive experiments are straightforward: compare adaptive versus uniform perception, hierarchical versus flat memory, ledger-constrained versus free-form generation, and one-pass versus two-pass operation under matched compute. If those experiments improve supported-claim precision, salient-event recall, identity continuity, and duration scaling, AV-Atlas will establish a defensible basis for a full long-form audiovisual intelligence platform.

Appendix A. Canonical event-ledger schema

The schema below is intentionally compact. Production records would add provenance IDs, model versions, language, geometry, relation graphs, and access-control fields. Dialogue text should retain its source (embedded subtitle, human transcript, or ASR) and must not be silently rewritten.

```
{
  "event_id": "EVT_0001842",
  "revision": 3,
  "start_ms": 4404080,
  "end_ms": 4410420,
  "level": "atomic",
  "scene_id": "SCN_0084",
  "claims": [
    {
      "claim_id": "CLM_01",
      "type": "visual_action",
      "text": "CHAR_07 closes a newspaper and looks toward the doorway.",
      "confidence": 0.93,
      "evidence_refs": ["VID:frame:18422-18574"]
    },
    {
      "claim_id": "CLM_02",
      "type": "non_speech_audio",
      "text": "Train brakes squeal outside.",
      "confidence": 0.78,
      "evidence_refs": ["AUD:4405.1-4408.6"]
    }
  ],
  "speech": [
    {
      "speaker_id": "CHAR_03",
      "text": "I thought you had already left.",
      "source": "asr",
      "word_alignment_ref": "ASR:turn:003889",
      "confidence": 0.96
    }
  ],
  "entities": ["CHAR_03", "CHAR_07", "OBJECT_NEWSPAPER"],
  "uncertainty": ["The item in CHAR_03's left hand may be a ticket envelope."],
  "status": "final",
  "provenance": {"source_id": "SRC_0017", "chunk_id": "CHK_0025"}
}
```

Appendix B. Experiment reporting template

Table B1. Minimum experiment report.

Field	Required report
Model identity	Backbone, encoders, trainable parameter count, context/memory limits, licenses.
Training data	Rights tiers, hours, domains, languages, annotation density, excluded data, contamination checks.
Training recipe	Stages, steps/tokens, batch, precision, optimizer, losses/weights, sampling curriculum, hardware.
Inference profile	Frame/audio policy, token budget, chunk/overlap, live latency, retrospective settings.
Automatic results	Component and end-to-end metrics with confidence intervals and duration strata.
Human results	Protocol, annotator count, agreement, adjudication, claim precision and event recall.
Ablations	At minimum A1-A7 from Table 14 at matched compute.
Efficiency	GPU-seconds/input-hour, peak memory, tokens/input-hour, storage, failure/retry rate.
Error taxonomy	Omission, hallucination, timing, identity, OCR, speech, audio, chronology, schema.
Governance	Provenance completeness, deletions, memorization tests, privacy/security findings.
Claim boundary	Explicit separation of measured results, targets, and qualitative observations.

Appendix C. Model and data release checklist

Table C1. Release checklist.

Area	Release question
Intended use	Which sources, durations, languages, and operating profiles were evaluated?
Out-of-scope use	Which identity, surveillance, legal, or high-stakes uses are unsupported or prohibited?
Weights and dependencies	Are all code, checkpoints, tokenizers, and alignment/diarization models licensed for the release?
Training content	Is a sufficiently detailed source-category summary available? Are rights reservations and exclusions documented?
Privacy	Can a source or person be deleted? Are embeddings and logs access controlled?
Evidence	Can every output claim expose the source window, model version, and confidence?
Performance	Are results stratified by duration, domain, language, modality, and source quality?
Memorization	Were exact/near-exact extraction and rare-sequence canaries tested?
Security	Was media-borne prompt injection tested? Are content and control channels isolated?
Operations	Can a ten-hour session recover from restart, chunk loss, and clock discontinuity?
Limitations	Are omissions, hallucinations, identity errors, and calibration limits described with examples?

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End of concept paper.